DSCI9001 Foundations of Data Practice: Reproducible Analysis

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Q1. Reproducibility practices

[Demo placeholder answer, written to test the pipeline rather than to model a good answer.] Reproducibility means another analyst can re-run an analysis and obtain the same result (DSCI9001 Teaching Team, 2026).

Environment pinning. Analyses often fail to reproduce because the computational environment is not recorded (Nguyen, 2024). Lock files address this by recording package versions, but lock files alone do not capture operating-system libraries (Nguyen, 2024), so containers may be needed. Containers close that gap but add build and maintenance overhead, so for a short analysis a lock file plus a recorded operating-system version is often the proportionate choice.

Version control. Version control identifies exactly which code produced a result (Nguyen, 2024). Its limitation is that it does not track large data files well, so data versions must be managed separately.

Neither practice is sufficient alone (Nguyen, 2024). For data analyses, environment pinning matters most, because unrecorded environments are the commonest cause of failure, but it only pays off when combined with version control.

Q2. Smoothing a sensor series

[Demo placeholder answer.] I implemented a centred moving average in `work/code/smoother.py` (`moving_average`, `rmse`). The edges are handled by shrinking the window, so the output keeps its length. All four supplied tests pass.

Table 1: RMSE between the smoothed and reference series.

Window	RMSE vs reference
raw (1)	0.789
3	0.517
7	0.426

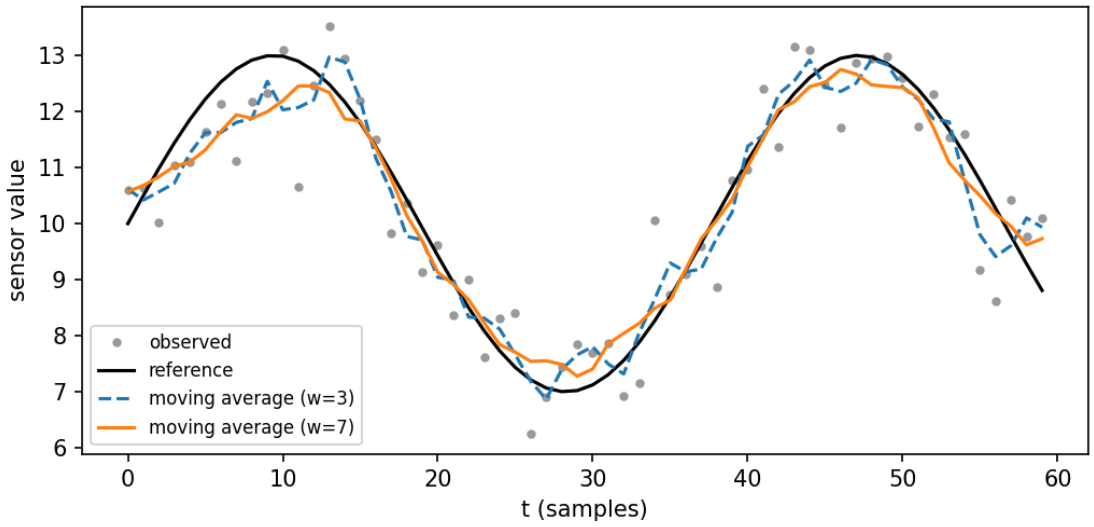


Figure 1: Observed, reference and smoothed series (w = 3, 7).

The wider window lowers the error by averaging out more noise, but it also flattens peaks. With a more rapidly varying signal it would add bias, so window size trades variance against bias.

AI use statement

[Demo text, generated from work/ai-use-log.md.] I used an AI coding agent through a command-line tool. It extracted the assignment requirements, built a marking checklist, planned both answers, drafted the text of Q1 and Q2 for me to revise, helped implement and test the smoother in Q2, and gave rubric-based feedback on my drafts. I verified every quotation against the set reading, re-ran all code myself, and rewrote the drafts in my own words. This use is consistent with the unit's policy on generative AI (Assignment 2 brief, p. 1).

References

DSCI9001 Teaching Team (2026) "Week 3: Reproducible workflows [lecture slides]."
Nguyen, A. (2024) "Practical reproducibility for data analysts."